



COMPETENCY STANDARD

FOR

IT SUPPORT SERVICE

(ICT SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for IT Support Service serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored for the ICT sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical for IT Support Service. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (20 hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-ICT-ISS-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-ICT-ISS-02-G	Carryout Workplace Interaction in English	1. Interpret workplace communication and etiquette 2. Read and understand workplace documents 3. Participate in workplace meetings and discussions 4. Practice professional ethics at workplace	10
Total Hour			20

Sector Specific Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-ICT-ISS-01-S	Operate Office Application Software	1. Operate computer 2. Install application software 3. Use word processor to prepare/create documents 4. Use spreadsheet to create/prepare worksheets 5. Use presentation software to create/prepare presentation	20
SICIP-ICT-ISS-02-S	Comply with Ethical Standards in IT Workplace	1. Uphold the requirements of clients 2. Deliver quality products and services 3. Maintain professionalism at workplace 4. Maintain workplace code of conduct	10
Total Hour			30

Occupation Specific Competencies (310 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-ICT-ISS-01-O	Interpret IT Fundamentals of Support Services	1. Interpret concepts of IT support services 2. Interpret IT devices and infrastructure 3. Interpret Troubleshooting Techniques in IT Systems 4. Interpret customer support services in IT system	20
SICIP-ICT-ISS-02-O	Carry out PC Assembling	1. Prepare for assemble 2. Disassemble hardware components of system unit 3. Assemble hardware components 4. Modify BIOS setting 5. Store and clean tools and equipment	50
SICIP-ICT-ISS-03-O	Install and Configure Operating Systems and Application Software	1. Install operating systems (OS). 2. Install required driver. 3. Install application and utility software. 4. Optimize system performance. 5. Update software and utilities.	50
SICIP-ICT-ISS-04-O	Manage Network and Internet Connectivity	1. Set up local area network (LAN) and Wi-Fi connections. 2. Configure network devices. 3. Monitor network performance. 4. Apply troubleshooting methods for network failures. 5. Secure network access and antivirus tools.	60
SICIP-ICT-ISS-05-O	Maintain Data Backup and Recovery Systems	1. Identify critical data and establish backup schedules. 2. Perform data backup using appropriate storage methods (cloud or local). 3. Test backup integrity to ensure data. 4. Recover lost or corrupted data. 5. Implement disaster recovery procedures.	30
SICIP-ICT-ISS-06-O	Implement Cybersecurity Measures	1. Identify security risks and system vulnerabilities. 2. Apply antivirus, anti-malware, and firewall	30

		protections. 3. Configure user access controls and permissions. 4. Monitor system and network activity for threats.	
SICIP-ICT-ISS-07-O	Maintain and Troubleshoot Personal Computer & Laptop	1. Inspect hardware components for faults. 2. Clean internal and external parts. 3. Diagnose hardware and software issues. 4. Repair or replace faulty components. 5. Install updates, drivers, and patches. 6. Test systems to ensure functionality. 7. Document issues and solutions for future reference.	70
Total Hour			310 Hrs.

COMPETENCY STANDARD: IT SUPPORT SERVICE

The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-ICT-ISS-01-G
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety procedures, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety procedures	2.1 OHS policies and procedures are practiced and applied. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal hygiene is practiced.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 First aid procedure is applied during emergency situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. OHS policies	1.1 International/ Local OHS requirements 1.2 Bangladesh Standards for OHS Building Code 1.3 Fire Safety Rules and Regulations
2. Personal Protective	2.1 Apron

Equipment (PPE)	2.2 Gas Mask 2.3 Gloves 2.4 Safety shoes 2.5 Face mask 2.6 Goggles 2.7 Ear plugs 2.8 Scarf
3. Hazards and risks	3.1 Chemical hazards 3.2 Biological hazards 3.3 Physical Hazards 3.4 Machine hazards 3.5 Materials hazards 3.6 Tools and Equipment hazards
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures 1.2 Work safety procedures 1.3 Emergency procedures – firefighting, earthquake response, explosion response & accident response 1.4 Types of hazards – biological, chemical & physical, and their effects 1.5 PPE types and uses 1.6 Personal hygiene practices 1.7 OHS awareness 1.8 Malfunctions and resolutions
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Applying personal work safety practices in the workplace 2.3 Identifying, assessing and controlling hazards and risks 2.4 Reporting incidents arising from hazards and risks 2.5 Responding to emergency procedures 2.6 Maintaining physical well-being in the workplace 2.7 Implementing emergency response plans and procedures 2.8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace

4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits 4.6 Learning manual
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 applied OHS policies and procedures 1.2 selected and used personal protective equipment (PPE) 1.3 practiced personal health and safety procedures 1.4 identified, assessed and controlled hazards and risks 1.5 reported incidents arising from hazards and risks to the authority 1.6 implemented plans and procedures to respond emergencies 1.7 applied basic first aid procedure
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT WORKPLACE INTERACTION IN ENGLISH	Nominal Duration: 10 hrs.	Unit Code: SICIP-ICT-ISS-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out workplace interaction in English. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions, and practicing professional ethics at workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace code of conducts are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a <u>courteous manner</u> to gather and convey information. 1.4 Questions about routine <u>workplace procedures and matters</u> are asked and responded as required.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted as per standard. 2.2 Assistance is taken to aid comprehension when required from peers/supervisors. 2.3 Visual information/symbols/signages are understood and followed. 2.4 Specific and relevant information are accessed from <u>appropriate sources</u> . 2.5 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time and followed meeting procedures and etiquette. 3.2 Own opinions are expressed and listened to those of others without interruption. 3.3 Inputs are provided consistent with the meeting purpose and interpreted and implemented meeting outcomes.
4. Practice professional ethics at workplace	4.1 Responsibilities as a team member are demonstrated and kept promises and commitments made to others. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is respected and maintained. 4.4 Situations and actions considered inappropriate or which present a conflict of interest are avoided.

Range of Variables

Variable	Range May Include but not limited to:
1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills
2. Workplace procedures and matters	2.1 Notes 2.2 Agenda 2.3 Simple reports such as progress and incident reports 2.4 Job sheets 2.5 Operational manuals 2.6 Brochures and promotional material 2.7 Visual and graphic materials 2.8 Standards 2.9 OSH information 2.10 Signs
3. Appropriate sources	3.1 HR Department 3.2 Managers 3.3 Supervisors

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Workplace code of conducts 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette
2. Underpinning Skills	2.1 Demonstrating performance of workplace communication and etiquette 2.2 Applying workplace information and symbols/signages 2.3 Applying workplace code of conducts as per organizational guidelines 2.4 Providing inputs in the meeting consistent with the meeting purpose 2.5 Interpreting and implementing meeting outcomes 2.6 Practicing professional ethics at workplace 2.7 Avoiding tactfully situations and actions considered inappropriate or which presents a conflict of interest
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Work book 4.3 Learning & operational manuals 4.4 Workplace tools/equipment 4.5 Relevant tools, equipment, software and facilities needed to perform the activities
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted workplace communication and etiquette 1.2 practiced workplace code of conducts as per organizational guidelines 1.3 Interpreted workplace documents, instructions and symbols 1.4 participated in workplace meetings and discussions 1.5 interpreted and implemented meeting outcomes 1.6 practiced professional ethics at workplace
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

The Sector Specific Competencies

Unit of Competency: OPERATE OFFICE APPLICATION SOFTWARE	Nominal Duration: 20 hrs.	Unit Code: SICIP-ICT-ISS-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate office application software. It specifically includes the tasks of operating computer, installing application software, using word processor to prepare/create documents, using spreadsheet to create/prepare worksheets and using presentation software to create/prepare presentation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Operate computer	1.1 Safe workplace practices are observed according to IT workplace guideline. 1.2 Desktop <u>Peripherals</u> are checked and connected with computer properly. 1.3 Computer is switched on. 1.4 Computer <u>desktop/GUI settings</u> are arranged and customized as per requirement. 1.5 Files and folders are <u>manipulated</u> as per requirement. 1.6 Properties of files and folders are viewed and searched. 1.7 Disks are defragmented, formatted as per requirement.
2. Install application software	2.1 Installation requirements of software are identified and listed. 2.2 Software sources and CD key/ password are assured. 2.3 <u>Appropriate Software</u> are collected and selected as per requirement. 2.4 Software installation is started. 2.5 Customization is done as per requirement. 2.6 Steps of installation are followed as per installation Instructions. 2.7 Installations are completed properly. 2.8 Correctness of Installation is checked.
3. Use word processor to prepare/create documents	3.1 Appropriate <u>word processor</u> is selected and started. 3.2 Documents are created as per requirement in personal use and office environment. 3.3 Contents are entered. 3.4 Documents are formatted. 3.5 Paragraph and page settings are completed. 3.6 Document is saved.
4. Use spreadsheet to create/prepare	4.1 <u>Spreadsheet applications</u> are selected and started. 4.2 Worksheets are created as per requirement in personal

worksheets	use and office environment. 4.3 Data are entered. 4.4 Functions are used for calculating and editing logical operation. 4.5 Sheets are formatted as per requirement. 4.6 Charts are created. 4.7 Charts/sheets are saved.
5. Use presentation software to create/prepare presentation	5.1 Appropriate <u>presentation applications</u> are selected and started. 5.2 Presentations are created as per requirement in personal use and office environment 5.3 Image, illustrations, text, table, symbols and media are entered as per requirements. 5.4 Presentations are formatted and animated. 5.5 Presentations are viewed and saved.

Range of Variables

Variable	Range
	May include but not limited to:
1. Peripherals	1.1 Monitor 1.2 Keyboard 1.3 Mouse 1.4 Modem 1.5 Scanner 1.6 Printer
2. Desktop settings	2.1 Icons 2.2 Taskbar 2.3 View 2.4 Resolutions
3. Manipulate	3.1 Create 3.2 Open 3.3 Copy 3.4 Rename 3.5 Delete 3.6 Sort
4. Typing tutors	4.1 English typing tutor 4.2 Bangla typing tutor
5. Appropriate software	5.1 MS office or Open office but limited to 5.2 Word processor software. 5.3 Spread sheet software 5.4 Presentation software
6. Word processor	6.1 MS Word processor 6.2 Open office Org 6.3 Google docs 6.4 Word perfect 6.5 LibreOffice
7. Spread sheet	7.1 MS Excel

applications	7.2 Google Sheets 7.3 Apple Numbers by Apple
8. Presentation application	8.1 MS PowerPoint 8.2 Google Slides 8.3 Prezi

Curricular Content Guide

1. Underpinning Knowledge	1.1 Desktop items 1.2 Type of Bangla keyboard layout 1.3 Different types of software and application packages 1.4 Use of word processor, spread sheet and presentation software 1.5 Types of printers 1.6 Types of charts, impotence of chart 1.7 Different types of math and logical functions
2. Underpinning Skills	2.1 Starting computer 2.2 Installing operating system 2.3 Managing desktop item 2.4 Manipulating files and folders as per requirement 2.5 Installing application software 2.6 Running application software 2.7 Creating and saving document with word processing application 2.8 Using functions in spread sheet. 2.9 Applying animations into presentation slide 2.10 Printing document
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Scanner, printer 4.4 Software 4.5 Work books 4.6 Relevant tools, equipment and facilities needed to perform the activities 4.7 Required learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate:
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	1.1 installed operating system 1.2 manipulated files and folders as per requirement 1.3 Installed appropriate application software 1.4 used word processor to prepare/create documents 1.5 used spreadsheet to create/prepare worksheets 1.6 Used presentation software to create/prepare presentation 1.7 applied animations into presentation slides 1.8 printed document
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: COMPLY WITH ETHICAL STANDARDS IN IT WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-ICT-ISS-02-S
Unit Descriptor: This unit covers knowledge, skills and attitudes required to comply with ethical standards in IT workplace. It specifically includes the tasks of upholding the requirements of clients, delivering quality products and services, maintaining professionalism at workplace and maintaining workplace code of conduct.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Uphold the requirements of clients	1.1 Clients' requirements are identified. 1.2 Confidentiality of information is maintained in accordance with workplace policies/organizational policies/national legislation. 1.3 Potential conflicts of interest are identified and involved parties of potential conflicts are notified. 1.4 Proprietary rights of client/customer are asserted.
2. Deliver quality products and services	2.1. Products and services are provided according to the clients' requirements. 2.2. Work is completed as per standards. 2.3. Quality processes are implemented when developing products and services.
3. Maintain professionalism at workplace	3.1 Work processes are delivered as per standards. 3.2 Skills, knowledge and qualifications are presented in a professional manner. 3.3 Services and products developed by self and others are delivered as per workplace standard. 3.4 Unbiased and objective information are provided to the clients. 3.5 Realistic estimates for time, cost and delivery of outputs are presented during negotiation.
4. Maintain workplace code of conduct	4.1 Workplace code of conduct is interpreted. 4.2 Workplace code of conduct is followed.

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Corporate code of confidentiality of information 1.2 Organizational policies, national legislation and workplace policies in relation to IT sector 1.3 Law and regulations pertaining to proprietary rights 1.4 Quality processes for products and services 1.5 Procedure of provided to client information 1.6 Method of estimating for time, cost and delivery products
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	and services 1.7 Workplace code of conduct in IT sector
2. Underpinning Skills	2.1 Upholding confidentiality of information in accordance with organizational policies, national legislation and workplace policies 2.2 Asserting proprietary rights of clients/customers 2.3 Completing work in accordance with industry and international standards 2.4 Implementing quality processes when developing products and services 2.5 Delivering correctly services and products developed by self and others 2.6 Providing unbiased and objective information to the clients 2.7 Presenting realistic estimates for time, cost and delivery of outputs during negotiation 2.8 Maintaining workplace code of conduct
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 National legislation in relation to IT sector 4.4 Workplace policies, code of conduct & proprietary rights of clients in IT sector 4.5 Relevant tools, equipment and facilities needed to perform the activities 4.6 Required learning materials

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 upheld the requirements of clients/customers 1.2 asserted proprietary rights of clients/customers 1.3 delivered quality products and services 1.4 Implemented quality processes when developing products and services 1.5 maintained professionalism at the workplace 1.6 completed work according to industry and international standards 1.7 maintained workplace code of conduct
2. Methods of	Methods of assessment should include:

Assessment	2.5 Written test 2.6 Practical demonstration 2.7 Oral question 2.8 Portfolio (optional)
3. Context of Assessment	3.3 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.4 Assessment should be done by a certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: INTERPRET IT FUNDAMENTALS OF SUPPORT SERVICES	Nominal Duration: 20 hrs.	Unit Code: SICIP-ICT-ISS-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to interpret IT fundamentals of support services. It specifically includes the tasks of interpreting concepts of IT support services, interpreting IT devices and infrastructure, interpret troubleshooting techniques in IT systems and interpreting customer support services in IT system.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret concepts of IT support services	1.1 Basic concepts of IT support services are interpreted. 1.2 Different types of <u>IT support services</u> are recognized. 1.3 <u>The roles and responsibilities of IT support personnel</u> are identified. 1.4 Common IT support processes are interpreted.
2. Interpret IT devices and infrastructure	2.1 <u>IT devices</u> are identified and interpreted. 2.2 Functions of <u>IT infrastructure components</u> are recognized. 2.3 Setup and configuration of IT devices are understood. 2.4 Relationship between different IT infrastructure components is interpreted. 2.5 Common issues affecting IT devices and infrastructure are identified.
3. Interpret Troubleshooting Techniques in IT Systems	3.1 <u>Common IT system issues</u> are identified. 3.2 Root causes of system issues are analyzed. 3.3 Standard troubleshooting techniques are interpreted. 3.4 Step-by-step problem-solving procedures are followed. 3.5 Preventive measures are recognized.
4. Interpret customer support services in IT system	4.1 Customer queries and issues are identified. 4.2 <u>Support procedures and protocols</u> are interpreted. 4.3 <u>Communication methods</u> are understood. 4.4 Common challenges in customer support are recognized.

Range of Variables

Variable	Range
	May include but not limited to:
1. IT support services	1.1 Hardware support 1.2 Software support

	1.3 Network support
2. The roles and responsibilities of IT support personnel	Roles: 2.1 Install and set up computers and software 2.2 Troubleshoot hardware/software issues 2.3 Maintain and upgrade systems 2.4 Ensure data backups and recovery 2.5 Monitor system performance and security 2.6 Provide user support and training Responsibilities: 2.7 Answer and resolve computer-related problems 2.8 Install updates, patches, and antivirus 2.9 Replace or repair faulty hardware 2.10 Keep records of issues and fixes 2.11 Protect computers from malware and unauthorized access
3. IT devices	3.1 Laptop 3.2 Desktop 3.3 Tablet 3.4 Smartphone 3.5 Server 3.6 Router 3.7 Switch 3.8 Firewall 3.9 Printer 3.10 Scanner 3.11 Storage drive (HDD/SSD/NAS) 3.12 Monitor 3.13 Keyboard 3.14 Mouse 3.15 UPS (power backup)
4. IT infrastructure components	4.1 Servers 4.2 Storage 4.3 Network (switches, routers, firewalls) 4.4 Data center / Cloud 4.5 Endpoints (PCs, laptops, phones) 4.6 OS & Software 4.7 Virtualization / Containers 4.8 Backup & Recovery 4.9 Security tools (antivirus, IAM, monitoring) 4.10 Internet / Connectivity
5. Common IT system issues	5.1 Network down 5.2 Slow internet 5.3 Server crash 5.4 Power outage 5.5 Hardware failure 5.6 Software bug 5.7 Login issues

	5.8 Virus / Malware 5.9 Data loss 5.10 Backup fail 5.11 Email not working 5.12 Printer jam / offline 5.13 Storage full 5.14 Update fail 5.15 Operating and pp crash
6. Support procedures and protocols	6.1 Remote support 6.2 On-site support 6.3 Troubleshooting steps 6.4 Escalation 6.5 Password reset 6.6 User account setup 6.7 Software install/update 6.8 Hardware replacement 6.9 Backup check/restore 6.10 Patch update 6.11 Documentation
7. Communication methods	7.1 Verbal 7.2 Written 7.3 Visual 7.4 Phone 7.5 Email 7.6 Chat

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify basic concepts, scope, and importance of IT support services. 1.2 Understand the roles and responsibilities of IT support personnel in maintaining systems and assisting users. 1.3 Interpret different types of IT devices, their functions, and their roles within IT infrastructure. 1.4 Recognize how various IT infrastructure components are connected and work together. 1.5 Understand common setup, configuration, and maintenance procedures for IT devices. 1.6 Identify frequent technical issues and interpret standard troubleshooting techniques. 1.7 Understand the steps for diagnosing root causes and applying effective solutions. 1.8 Recognize preventive measures to avoid recurring IT problems. 1.9 Interpret the importance of effective customer support, communication skills, and standard service procedures
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	in IT operations.
2. Underpinning Skills	2.1 Performing identification and interpretation of basic IT support concepts. 2.2 Demonstrating recognition of different types of IT support services. 2.3 Following standard IT support processes and procedures. 2.4 Applying setup and configuration techniques for IT devices and infrastructure components. 2.5 Performing troubleshooting to detect and analyze common IT system issues. 2.6 Following step-by-step problem-solving procedures to resolve system faults. 2.7 Applying preventive measures to avoid recurring IT problems. 2.8 Demonstrating effective communication and customer support skills in IT service delivery. 2.9 Following protocols for handling customer queries, issues, and feedback efficiently.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Relevant tools 4.4 Facilities needed to perform the activities 4.5 Required learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 interpreted concepts of IT support services 1.2 interpreted IT devices and infrastructure 1.3 interpreted troubleshooting techniques in IT systems 1.4 interpreted customer support services in IT system
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration

	2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT PC ASSEMBLING	Nominal Duration: 50 hrs.	Unit Code: SICIP-ICT-ISS-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry out PC assembling. It specifically includes the tasks of preparing for assemble, disassembling hardware components of system unit, assembling hardware components, modifying BIOS setting and storing and cleaning tools and equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for assemble	1.1 <u>Personal Protective Equipment (PPE)</u> is collected and used. 1.2 Work area is prepared, ensuring it is clean, organized, and free from static. 1.3 All necessary <u>tools and equipment</u> are gathered. 1.4 All PC components are identified and checked. 1.5 <u>Anti-static precautions</u> are applied to prevent component damage. 1.6 Component specifications are reviewed.
2. Disassemble hardware components of system unit	2.1 <u>System unit</u> is powered off and disconnected from all power sources. 2.2 <u>External components</u> are removed to provide clear access to the system unit. 2.3 Casing of the system unit is opened carefully to avoid damage to internal components. 2.4 <u>Internal hardware components</u> are removed following proper procedures. 2.5 Components are handled using anti-static precautions. 2.6 Disassembled components are organized and stored safely for future assembly or maintenance.
3. Assemble hardware components	3.1 Internal hardware components are installed in the system unit following proper procedures. 3.2 All components are aligned and secured ensuring stability and proper fit. 3.3 Connections are properly attached according to component specifications. 3.4 Anti-static precautions are applied throughout the assembly process. 3.5 Assembled system unit is checked to ensure that all components are correctly installed and ready for testing.
4. Modify BIOS setting	4.1 System is powered on and accessed to enter the BIOS setup utility safely.

	4.2 Existing BIOS settings are reviewed to understand current configurations for the system hardware. 4.3 Required BIOS settings are modified according to the hardware specifications. 4.4 Changes are saved and confirmed, ensuring that the system recognizes the updated configurations. 4.5 System is restarted and monitored to verify the BIOS modifications are done.
5. Store and clean tools and equipment	5.1 Tools and equipment are cleaned after use to remove dust, dirt, and residues from components and surfaces. 5.2 All tools are inspected for damage or wear, and any faulty tools are set aside for repair or replacement. 5.3 Tools and equipment are organized and stored in designated storage areas.

Range of Variables

Variable	Range May include but not limited to:
1. Personal Protective Equipment (PPE)	1.1 Apron 1.2 Musk 1.3 Anti-static mat 1.4 Anti-static shoe 1.5 Safety hand gloves 1.6 Safety goggles
2. Tools and equipment	2.1 PC Assemblers Tool box 2.2 ESD tools 2.3 AVO meter 2.4 Power extension cord with multiple power socket 2.5 Blower gun 2.6 Hand brush
3. Anti-static precautions	3.1 Wear an anti-static wrist strap (and clip it to ground). 3.2 Work on an anti-static mat. 3.3 Touch a grounded metal object before handling components. 3.4 Keep components in anti-static bags when not in use. 3.5 Avoid carpets (they build static fast). 3.6 Unplug the device before opening it up. 3.7 Handle parts by the edges, not the circuitry. 3.8 Keep humidity moderate (dry air = more static).
4. System unit	4.1 Motherboard 4.2 CPU 4.3 RAM 4.4 Power Supply (PSU) 4.5 Storage 4.6 GPU 4.7 Cooling system 4.8 Expansion cards

	4.9 Ports & connectors
5. External components	5.1 Monitor 5.2 Keyboard 5.3 Mouse 5.4 Printer/Scanner 5.5 Speakers/Headphones 5.6 External drives (HDD/SSD/USB) 5.7 Webcam 5.8 Microphone 5.9 Docking station/USB hub 5.10 Network devices
6. Internal hardware components	6.1 Motherboard 6.2 Processor 6.3 RAM 6.4 Storage (HDD/SSD) 6.5 GPU 6.6 PSU 6.7 Cooling system 6.8 Expansion cards – sound card, network card, etc. 6.9 Cables & connectors – power/data links
7. BIOS settings	7.1 Boot Order 7.2 UEFI / Legacy 7.3 Virtualization 7.4 XMP / DOCP (RAM) 7.5 CPU Ratio / Voltage 7.6 Fan Control 7.7 SATA Mode (AHCI/RAID) 7.8 Integrated Graphics

Curricular Content Guide

1. Underpinning Knowledge	1.1 Understand the procedures for preparing a clean, organized workspace before assembling computer components. 1.2 Recognize different PC components, their specifications, and functions. 1.3 Interpret proper methods for disassembling system units safely. 1.4 Understand anti-static precautions and their role in protecting sensitive hardware. 1.5 Identify correct procedures for assembling hardware components and ensuring proper alignment and connection. 1.6 Interpret BIOS functions, settings, and procedures for modifying configurations according to hardware requirements. 1.7 Understand how to verify and test assembled systems
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	for proper functionality. 1.8 Recognize the importance of cleaning, inspecting, and storing tools and equipment properly after work completion.
2. Underpinning Skills	2.1 Preparing and organizing work areas to ensure safety and efficiency. 2.2 Gathering and handling tools, equipment, and PC components appropriately. 2.3 Disassembling system units and internal hardware components following correct procedures. 2.4 Applying anti-static precautions during disassembly and assembly processes. 2.5 Assembling hardware components accurately, ensuring proper alignment, connection, and stability. 2.6 Modifying and configuring BIOS settings according to hardware specifications. 2.7 Testing and verifying system functionality after assembly and configuration. 2.8 Cleaning, inspecting, and storing tools and equipment safely after use.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Multimedia 4.4 Tools (content writing, graphic design, video creation) 4.5 Required learning materials 4.6 Specifications and work instructions

Assessment Evidence Guide

	Assessment must evidence that the candidate:
1. Critical Aspects of Competency	1.1 prepared for assemble 1.2 disassembled hardware components of system unit 1.3 assembled hardware components 1.4 modified BIOS setting 1.5 stored and cleaned tools and equipment
2. Methods of	Methods of assessment should include:

Assessment	2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INSTALL AND CONFIGURE OPERATING SYSTEMS AND APPLICATION SOFTWARE	Nominal Duration: 50 hrs.	Unit Code: SICIP-ICT-ISS-03-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to install and configure operating systems and application software. It specifically includes the tasks of installing operating systems (OS), installing required driver, installing application and utility software, optimizing system performance and updating software and utilities.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Install operating system (OS).	1.1 System hardware is prepared and checked. 1.2 Installation <u>media</u> for the <u>operating system</u> is selected and loaded onto the system. 1.3 Operating system is installed according to the manufacturer's guidelines and installation procedures. 1.4 Partitioning and formatting of storage devices are performed. 1.5 Post-installation configurations are applied.
2. Install required driver.	2.1 <u>Required drivers</u> are identified based on the operating system and connected hardware components. 2.2 Driver installation files are obtained from official or <u>verified sources</u> to ensure authenticity and compatibility. 2.3 Drivers are installed following the manufacturer's guidelines and installation procedures. 2.4 Installed drivers are configured. 2.5 Functionality of the hardware components is tested after driver installation to confirm proper operation.
3. Install application and utility software.	3.1 Required <u>application software</u> and <u>utility software</u> are identified based on the system and user requirements. 3.2 Software installation files are obtained from trusted and verified sources. 3.3 Software is installed according to the manufacturer's guidelines and installation procedures. 3.4 Post-installation configurations are applied. 3.5 Functionality of the installed software is tested.
4. Optimize system performance.	4.1 System performance is assessed by monitoring key parameters. 4.2 Unnecessary applications and services are disabled or removed. 4.3 System updates and patches are applied.

	4.4 Storage and memory are managed to enhance system speed and efficiency. 4.5 Startup programs are configured to minimize boot time and improve overall system performance.
5. Update software and utilities.	5.1 Installed software and utilities are checked for available updates and new versions. 5.2 Updates are downloaded from trusted and verified sources. 5.3 Software and utilities are updated following the manufacturer's instructions and standard procedures. 5.4 Functionality of the updated software is tested.

Range of Variables

Variable	Range
	May include but not limited to:
1. Media	1.1 Pen drive 1.2 Portable hard disk 1.3 DVD or CD
2. Operating system	2.1 MacOS 2.2 Windows 2.3 Linux
3. Required drivers	3.1 Chipset 3.2 Graphics (GPU) 3.3 Audio 3.4 LAN / Wi-Fi / Bluetooth 3.5 Printer / Scanner 3.6 Touchpad 3.7 Storage (SATA / NVMe / RAID) 3.8 USB Controller
4. Verified sources	4.1 Official Manufacturer Website (Dell, HP, Lenovo, etc.) 4.2 Component Vendor Website (Intel, AMD, NVIDIA, Realtek, Broadcom, etc.) 4.3 Microsoft Update Catalog 4.4 Windows Update
5. Application software	5.1 Microsoft Office 5.2 Typing software 5.3 Browsers 5.4 Google Workspace 5.5 Microsoft Teams 5.6 Zoom 5.7 FileZilla 5.8 Malwarebytes 5.9 Zendesk 5.10 Dropbox 5.11 OneDrive
6. Utility software	6.1 CCleaner 6.2 WinRAR / 7-Zip

	6.3 Disk Cleanup 6.4 Task Manager 6.5 PowerShell 6.6 Event Viewer 6.7 Windows Defender 6.8 Driver Booster
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify system hardware requirements and preparation steps for operating system installation. 1.2 Understand different types of operating systems, installation media, and standard installation procedures. 1.3 Interpret concepts of disk partitioning, formatting, and post-installation configurations. 1.4 Recognize the purpose and importance of hardware drivers for system functionality. 1.5 Understand proper methods to obtain, install, and configure drivers from verified sources. 1.6 Identify various types of application and utility software and their installation processes. 1.7 Interpret methods to test and verify software functionality after installation. 1.8 Understand techniques for optimizing system performance through updates, storage management, and startup configuration. 1.9 Recognize the importance of keeping software and utilities updated for system stability, security, and performance.
2. Underpinning Skills	2.1 Performing preparation and verification of system hardware before OS installation. 2.2 Following procedures to load installation media and install operating systems accurately. 2.3 Applying partitioning, formatting, and post-installation configurations on storage devices. 2.4 Configuring installed drivers and testing hardware functionality. 2.5 Installing application and utility software following standard procedures and verifying operation. 2.6 Assessing system performance and applying optimization techniques for speed and efficiency. 2.7 Applying updates, patches, and software upgrades according to manufacturer instructions. 2.8 Testing and verifying system and software functionality after updates and installations.

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Specifications and work instructions 4.4 Learning materials 4.5 Relevant equipment, software and facilities needed to perform the activities

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 installed operating systems (OS) 1.2 installed required driver 1.3 installed application and utility software 1.4 optimized system performance 1.5 updated software and utilities
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MANAGE NETWORK AND INTERNET CONNECTIVITY	Nominal Duration: 60 hrs.	Unit Code: SICIP-ICT-ISS-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to manage network and internet connectivity. It specifically includes the tasks of setting up local area network (LAN) and Wi-Fi connections, configuring network devices, monitoring network performance, applying troubleshooting methods for network failures and securing network access and antivirus tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Set up local area network (LAN) and Wi-Fi connections.	1.1 LAN requirements are identified. 1.2 Appropriate hardware devices are identified and selected and prepared according to specifications. 1.3 <u>Network cables</u> are installed, connected, and labeled as per standard procedures. 1.4 Wi-Fi router and access points are configured following manufacturer's guidelines. 1.5 <u>Network devices</u> are powered, tested, and verified for operational status. 1.6 Security settings and access restrictions are applied according to organizational policy. 1.7 Connectivity is confirmed and functionality is documented in line with workplace requirements.
2. Configure network devices.	2.1 Configuration requirements are obtained from organization. 2.2 Network devices are accessed using appropriate <u>interfaces</u> and methods. 2.3 Device settings are configured according to specifications and technical guidelines. 2.4 <u>Internet Protocol (IP)</u> addresses, subnet masks, and gateway parameters are assigned as required. 2.5 Routing, switching, and wireless settings are adjusted and saved as per instructions. 2.6 <u>Security features</u> are enabled in line with organizational standards. 2.7 Device configurations are tested and verified for correct operation.
3. Monitor network performance.	3.1 Network monitoring requirements are identified. 3.2 <u>Network monitoring tools</u> are selected and applied according to specifications. 3.3 Traffic patterns and <u>bandwidth</u> utilization are observed and recorded as instructed.

	3.4 Device logs and performance metrics are reviewed to detect abnormalities. 3.5 Performance data is analyzed and compared against baseline standards. 3.6 Monitoring reports are prepared and submitted according to workplace requirements.
4. Apply troubleshooting methods for network failures.	4.1 Network failure symptoms are identified from monitoring data and user reports. 4.2 Hardware and software components are checked to locate the source of the problem. 4.3 Fault isolation methods are applied according to standard troubleshooting procedures. 4.4 Configuration settings are verified and corrected as required. 4.5 Diagnostic tools are used to test connectivity and network performance. 4.6 Solutions are implemented and tested to confirm successful restoration of services.
5. Secure network access and antivirus tools.	5.1 Organizational security requirements are identified. 5.2 User access rights and permissions are configured according to authorization levels. 5.3 Network authentication methods are applied as instructed. 5.4 Firewalls and security rules are enabled to protect against unauthorized access. 5.5 Antivirus software is installed and updated according to vendor specifications. 5.6 Detected threats and malicious activities are quarantined or removed using approved procedures.

Range of Variables

Variable	Range
	May include but not limited to:
1. Network devices	1.1 Switches 1.2 Hubs 1.3 Routers 1.4 Network interface cards (NICs) 1.5 Access Point (AP) 1.6 Firewall Device 1.7 Modem 1.8 Patch panels
2. Network cables	2.1 UTP 2.2 STP 2.3 Fiber optic cables 2.4 Patch chords
3. Interfaces	3.1 Command line interface (CLI)

	3.2 Graphical user interface (GUI) 3.3 Web interface 3.4 Console cable
4. Internet Protocol (IP)	4.1 TCP/IP 4.2 DHCP 4.3 DNS 4.4 VLAN 4.5 NAT 4.6 routing protocols (RIP, OSPF, EIGRP).
5. Security features	5.1 Access control lists (ACLs) 5.2 Firewall rules 5.3 Port security 5.4 Encryption (WPA2/WPA3) 5.5 Strong password policies
6. Network monitoring tools	6.1 SNMP tools 6.2 Wireshark 6.3 PRTG 6.4 Nagios 6.5 SolarWinds 6.6 MRTG 6.7 Built-in device monitoring utilities
7. Bandwidth	7.1 Bandwidth usage 7.2 Latency 7.3 Jitter 7.4 Packet loss 7.5 Throughput 7.6 Uptime/downtime
8. Device logs	8.1 System logs 8.2 Event logs 8.3 Performance counters 8.4 Real-time dashboards
9. Software components	9.1 Firmware 9.2 Drivers 9.3 Operating system network settings 9.4 Firewall software
10. Fault isolation methods	10.1 Fault isolation 10.2 Step-by-step testing 10.3 Divide-and-conquer 10.4 Substitution method 10.5 Escalation procedures
11. Diagnostic tools	11.1 Ping 11.2 Traceroute 11.3 Ipconfig/ifconfig 11.4 Netstat 11.5 Cable tester 11.6 Wireshark 11.7 SNMP monitors

12. Antivirus software	12.1 Windows Defender 12.2 Kaspersky 12.3 McAfee 12.4 Avast 12.5 Norton 12.6 Enterprise endpoint security solutions
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify the basic concepts and requirements for setting up Local Area Networks (LAN) and Wi-Fi connections. 1.2 Understand the functions and specifications of network hardware devices. 1.3 Interpret standard procedures for installing, connecting, and labeling network cables. 1.4 Recognize key configurations for routers, access points, and wireless settings following manufacturer guidelines. 1.5 Understand network addressing concepts. 1.6 Interpret methods for configuring and securing network devices according to organizational standards. 1.7 Identify tools and techniques used for monitoring network traffic, bandwidth, and device performance. 1.8 Understand common troubleshooting methods for identifying and resolving network failures. 1.9 Recognize the importance of network security measures.
2. Underpinning Skills	2.1 Performing identification and preparation of LAN and Wi-Fi network requirements. 2.2 Selecting, installing, and configuring network hardware devices and cables according to specifications. 2.3 Applying manufacturer guidelines to set up routers, access points, and wireless connections. 2.4 Testing and verifying network connectivity and operational status of devices. 2.5 Configuring network settings including IP addresses, routing, switching, and wireless parameters. 2.6 Monitoring network performance using appropriate tools and analyzing traffic and device metrics. 2.7 Applying troubleshooting procedures to diagnose and resolve network failures. 2.8 Implementing security measures including user access controls, firewalls, and antivirus protection.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn

	3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Job specifications and work instructions 4.4 Relevant tools, equipment and facilities 4.5 Required learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 set up local area network (LAN) and Wi-Fi connections. 1.2 configured network devices. 1.3 monitored network performance. 1.4 applied troubleshooting methods for network failures. 1.5 secured network access and antivirus tools.
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MAINTAIN DATA BACKUP AND RECOVERY SYSTEMS	Nominal Duration: 30 hrs.	Unit Code: SICIP-ICT-ISS-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to maintain data backup and recovery systems. It specifically includes the tasks of identifying critical data and established backup schedules, performed data backup using appropriate storage methods (cloud or local), testing backup integrity to ensure data, recovering lost or corrupted data and implementing disaster recovery procedures.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify critical data and establish backup schedules.	1.1 Organizational requirements for data backup are identified. 1.2 <u>Backup methods</u> are selected according to organizational and technical requirements. 1.3 Backup schedules are developed as per workplace standards. 1.4 <u>Backup tools</u> and <u>storage media</u> are identified and prepared as specified.
2. Perform data backup using appropriate storage methods (cloud or local).	2.1 Backup requirements are identified. 2.2 Appropriate storage methods (cloud or local) are selected according to data type and security needs. 2.3 Backup tools are configured as specified. 2.4 <u>Data</u> is backed up using approved methods and verified for accuracy and completeness. 2.5 Backup files are secured and stored in line with organizational standards.
3. Test backup integrity to ensure data.	3.1 Organizational requirements for backup testing are identified. 3.2 Backup media and storage devices are accessed using approved methods. 3.3 Test restorations are performed to verify the accuracy and completeness of backup data. 3.4 Backup files are validated using <u>verification tools</u> and checksum <u>methods</u> . 3.5 Restored data is compared with original data to ensure integrity and consistency.
4. Recover lost or corrupted data.	4.1 Data recovery requirements are identified. 4.2 <u>Recovery tools</u> and <u>methods</u> are selected according to the type and extent of data loss.

	4.3 Recovery processes are applied to retrieve lost or corrupted files. 4.4 Data is restored to designated systems. 4.5 Restored data is verified for accuracy, completeness, and usability. 4.6 <u>Security measures</u> are applied during and after recovery to protect sensitive information.
5. Implement disaster recovery procedures	5.1 Disaster recovery requirements are identified. 5.2 <u>Disaster recovery plans</u> are reviewed and updated. 5.3 Recovery roles and responsibilities are assigned and communicated. 5.4 <u>Recovery procedures</u> are executed using approved methods and tools. 5.5 Critical systems and data are restored to operational status within required timeframes. 5.6 System functionality is tested and verified after recovery activities.

Range of Variables

Variable	Range May include but not limited to:
1. Backup methods	1.1 Full backup 1.2 Incremental backup 1.3 Differential backup 1.4 Cloud backup 1.5 Snapshot backup
2. Backup tools	2.1 Windows Backup 2.2 Veeam 2.3 Acronis 2.4 Commvault 2.5 Google Drive/OneDrive 2.6 Enterprise backup systems
3. Storage media	3.1 Local disks 3.2 External hard drives 3.3 NAS devices 3.4 Cloud storage 3.5 Tape drives
4. Data	4.1 Databases 4.2 Application files 4.3 Emails 4.4 Multimedia 4.5 Business-critical records
5. Verification tools	5.1 Backup software validation features 5.2 Third-party integrity checkers 5.3 Command-line utilities

6. Methods	6.1 Checksum/MD5/SHA validation 6.2 File comparison 6.3 Log file review 6.4 Test restore
7. Recovery tools	7.1 Recuva 7.2 EaseUS 7.3 Stellar Data Recovery 7.4 Ontrack 7.5 TestDisk 7.6 Built-in OS recovery utilities
8. Recovery methods	8.1 Data recovery software 8.2 Backup restoration 8.3 File system repair 8.4 RAID recovery 8.5 Cloud recovery tools 8.6 Professional recovery services
9. Security measures	9.1 Encryption 9.2 Access control 9.3 Antivirus scanning after recovery 9.4 Secure transfer protocols.
10. Disaster recovery plans	10.1 Business continuity plan (BCP) 10.2 Disaster recovery plan (DRP) 10.3 Emergency response plan
11. Recovery procedures	11.1 Backup restoration 11.2 System rebuilds 11.3 Cloud-based recovery 11.4 Failover procedures 11.5 Redundant systems utilization

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify the importance of data backup and understand organizational requirements for data protection. 1.2 Interpret different backup methods and storage options, including local and cloud-based systems. 1.3 Understand how to develop and maintain effective backup schedules in line with workplace standards. 1.4 Recognize tools, storage media, and software used for performing secure and reliable backups. 1.5 Interpret methods for testing backup integrity through verification tools and test restorations. 1.6 Understand key procedures for recovering lost or corrupted data using appropriate recovery tools. 1.7 Identify techniques to ensure accuracy, completeness, and security of restored data. 1.8 Understand the purpose and structure of disaster recovery plans and how they support business
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	continuity. 1.9 Recognize the roles, responsibilities, and processes involved in restoring systems and data after a disaster.
2. Underpinning Skills	2.1 Performing identification of critical data and organizational backup requirements. 2.2 Selecting and applying appropriate backup methods and storage media (cloud or local). 2.3 Configuring backup tools and performing data backups accurately and securely. 2.4 Testing backup integrity using verification tools and performing test restorations. 2.5 Comparing restored data with original data to ensure accuracy, completeness, and consistency. 2.6 Applying data recovery procedures to retrieve lost or corrupted files. 2.7 Implementing security measures during backup and recovery processes to protect sensitive information. 2.8 Executing disaster recovery procedures to restore critical systems and data within required timeframes. 2.9 Testing and verifying system functionality after recovery activities to ensure operational readiness.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Learning materials 4.4 Specifications and work instructions 4.5 Relevant tools, equipment and facilities

Assessment Evidence Guide

	Assessment must evidence that the candidate:
1. Critical Aspects of Competency	1.1 Identified critical data and established backup schedules 1.2 performed data backup using appropriate storage methods (cloud or local) 1.3 tested backup integrity to ensure data 1.4 recovered lost or corrupted data

	1.5 implemented disaster recovery procedures
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IMPLEMENT CYBERSECURITY MEASURES	Nominal Duration: 30 hrs.	Unit Code: SICIP-ICT-ISS-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to implement cybersecurity measures. It specifically includes the tasks of identifying security risks and system vulnerabilities, applying antivirus, anti-malware and firewall protections, configuring user access controlling and permissions, monitoring system and network activity for threats.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify security risks and system vulnerabilities.	1.1 Security policies and requirements are reviewed and interpreted. 1.2 System components, applications, and network devices are inspected to detect potential vulnerabilities. 1.3 <u>Security risks</u> are assessed using approved risk identification <u>methods</u> and <u>tools</u> . 1.4 Threats, weaknesses, and possible points of compromise are identified. 1.5 Vulnerability scans and tests are conducted according to organizational procedures. 1.6 Findings are analyzed and prioritized based on potential impact and likelihood.
2. Apply antivirus, anti-malware, and firewall protections.	2.1 <u>Antivirus</u> and <u>anti-malware</u> software are installed and updated according to vendor and workplace guidelines. 2.2 Firewall settings are configured as per organizational standards and security requirements. 2.3 System scans and threat detection are scheduled and executed using approved methods. 2.4 Detected threats and vulnerabilities are quarantined, removed, or mitigated following approved procedures.
3. Configure user access controls and permissions.	3.1 <u>User accounts</u> are created, modified, or disabled. 3.2 <u>Access permissions</u> are assigned according to roles, responsibilities, and authorization levels. 3.3 <u>Authentication methods</u> are applied in line with organizational security standards. 3.4 Access restrictions and privileges are tested. 3.5 User access activity is monitored and logged to detect unauthorized attempts.
4. Monitor system and network activity for	4.1 <u>Monitoring tools</u> are selected and applied according to specifications.

threats.	4.2 System and network activity logs are collected and reviewed regularly. 4.3 Alerts and notifications of suspicious activity are received, acknowledged, and recorded. 4.4 Potential <u>security threats</u> and vulnerabilities are identified through analysis of activity data. 4.5 Detected threats are escalated and corrective actions are applied. 4.6 Monitoring results, incidents, and actions taken are documented.
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Range of Variables

Variable	Range May include but not limited to:
1. Security risks	1.1 Outdated software 1.2 Misconfigured devices 1.3 Weak passwords 1.4 Unpatched systems 1.5 Open ports 1.6 Malware 1.7 Phishing threats
2. Methods	2.1 Vulnerability scanning 2.2 Penetration testing 2.3 Configuration audits 2.4 Security checklists 2.5 Threat modeling.
3. Tools	3.1 Nessus 3.2 OpenVAS 3.3 Qualys 3.4 Nmap 3.5 Wireshark 3.6 Built-in OS security tools
4. Antivirus software	4.1 Kaspersky 4.2 Norton 4.3 McAfee
5. Anti-malware software	5.1 Malwarebytes 5.2 Firewalls 5.3 Endpoint security suites
6. User accounts	6.1 New users 6.2 Existing users 6.3 Temporary accounts 6.4 Administrative accounts 6.5 Guest accounts
7. Access permissions	7.1 Read 7.2 Write 7.3 Execute

	7.4 Delete 7.5 Modify 7.6 Administrative privileges
8. Authentication methods	8.1 Passwords 8.2 Security tokens 8.3 Smart cards 8.4 OTP 8.5 Biometric verification
9. Monitoring tools	9.1 SIEM tools 9.2 Intrusion detection/prevention systems (IDS/IPS) 9.3 Antivirus/antimalware dashboards 9.4 Firewall logs 9.5 Network monitoring software
10. Security threats	10.1 Malware 10.2 Ransomware 10.3 Unauthorized access attempts 10.4 Phishing attacks 10.5 Suspicious traffic patterns 10.6 Abnormal system activity

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify key concepts of information security, system vulnerabilities, and organizational security policies. 1.2 Understand methods and tools used to assess and identify security risks within systems and networks. 1.3 Interpret procedures for conducting vulnerability scans, analyzing findings, and prioritizing threats. 1.4 Recognize the functions of antivirus, anti-malware, and firewall tools in protecting system integrity. 1.5 Understand how to configure and maintain security software according to organizational standards. 1.6 Identify the principles of user access control, authentication, and role-based permissions. 1.7 Interpret techniques for monitoring system and network activities to detect and respond to security threats. 1.8 Understand procedures for handling, reporting, and documenting security incidents. 1.9 Recognize the importance of maintaining up-to-date security measures to safeguard organizational data and systems.
2. Underpinning Skills	2.1 Performing identification and assessment of security risks and system vulnerabilities. 2.2 Applying antivirus, anti-malware, and firewall protections according to organizational standards. 2.3 Installing, configuring, and updating security software to prevent threats. 2.4 Configuring user accounts, access permissions, and

	authentication methods. 2.5 Monitoring system and network activity using appropriate tools to detect security threats. 2.6 Applying corrective actions to mitigate detected threats and vulnerabilities. 2.7 Documenting security incidents, monitoring results, and applied solutions. 2.8 Following organizational security policies and procedures to maintain system integrity.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Specifications and work instructions 4.4 Relevant tools, equipment and facilities 4.5 Required learning materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment must evidence that the candidate: 1.1 identified security risks and system vulnerabilities 1.2 applying antivirus anti malware and firewall protections 1.3 configuring user access controls and permissions 1.4 monitoring system and network activity for threats.
2. Methods of Assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MAINTAIN AND TROUBLESHOOT PERSONAL COMPUTER & LAPTOP	Nominal Duration: 70 Hrs.	Unit Code: SICIP-ICT-ISS-07-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to maintain and troubleshooting personal computer and laptop. It specifically includes the tasks of inspecting hardware components for faults, cleaning internal and external parts, diagnosing hardware and software issues, repairing or replacing faulty components, installing updates drivers and patches, testing systems to ensure functionality and documenting issues and solutions for future reference.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Inspect hardware components for faults.	1.1 <u>Tools and equipment</u> for inspection are selected and prepared as required. 1.2 <u>Hardware components</u> are examined for physical damage, wear, or malfunction. 1.3 Electrical connections, ports, and interfaces are tested to confirm proper operation. 1.4 <u>Faults</u> , anomalies, or deviations from specifications are identified and recorded.
2. Clean internal and external parts.	2.1 <u>Cleaning tools</u> , equipment, and materials are selected and prepared as required. 2.2 External parts of hardware and devices are cleaned using approved <u>methods</u> and materials. 2.3 Internal components are cleaned carefully to remove dust, debris, and contaminants without causing damage. 2.4 Cleaning activities are performed following <u>safety</u> and manufacturer guidelines. 2.5 Faults, damage, or irregularities detected during cleaning are reported to authorized personnel.
3 Diagnose hardware and software issues.	3.1 <u>Diagnostic tools</u> and software are selected and prepared according to specifications. 3.2 Hardware components are examined to detect faults, malfunctions, or performance issues. 3.3 <u>Software systems</u> are analyzed to identify errors, configuration problems, or compatibility issues. 3.4 Symptoms of hardware and software problems are <u>documented</u> and recorded. 3.5 Faults are isolated and prioritized based on severity and impact on operations.
4 Repair or replace faulty	4.1 Faulty <u>components</u> are isolated and confirmed using

components.	diagnostic tools and procedures. 4.2 <u>Repair methods</u> or <u>replacement parts</u> are selected according to manufacturer specifications. 4.3 Components are repaired or replaced using approved procedures and tools. 4.4 Functionality of repaired or replaced components is tested to ensure proper operation. 4.5 Safety measures are applied throughout the repair or replacement process.
5 Install updates, drivers, and patches.	5.1 Required <u>updates</u>, drivers, and patches are selected according to system type and compatibility. 5.2 <u>Installation tools</u> is used and procedures are followed according to manufacturer guidelines. 5.3 Updates, drivers, and patches are applied to operating systems, applications, and hardware components. 5.4 System functionality is verified after installation to ensure successful implementation.
6 Test systems to ensure functionality.	6.1 <u>Testing tools</u> and <u>procedures</u> are selected and prepared as required. 6.2 System components, hardware, and software are tested according to approved methods. 6.3 Functional tests are performed to verify proper operation, performance, and reliability. 6.4 Test results are compared against <u>expected outcomes</u> and system specifications. 6.5 Corrective actions are applied as instructed for identified issues.
7 Document issues and solutions for future reference.	7.1 Documentation requirements are identified. 7.2 <u>Issues</u> encountered during operations are clearly described and recorded in approved formats. 7.3 Root causes of issues are identified and documented according to diagnostic findings. 7.4 <u>Solutions</u> and corrective actions applied are recorded in detail for future use. 7.5 <u>Documentation</u> is reviewed and verified to ensure accuracy and completeness. 7.6 Records are stored securely and made accessible to authorized personnel.

Range of Variables

Variable	Range (May includes but not limited to):
1. Tools and equipment	1.1 Multi meter 1.2 Cable tester 1.3 Diagnostic software 1.4 Thermal camera, 1.5 Screwdrivers

	1.6 Anti-static equipment
2. Hardware components	2.1 Servers 2.2 Desktops 2.3 Laptops 2.4 Network devices 2.5 Storage devices 2.6 Peripherals
3. Faults	3.1 Physical damage 3.2 Overheating, 3.3 Loose connections 3.4 Corrupted components 3.5 Worn-out parts 3.6 Failing storage devices
4. Cleaning tools	4.1 Microfiber cloths 4.2 Soft brushes 4.3 Compressed air 4.4 Vacuum cleaners 4.5 Isopropyl alcohol 4.6 Cleaning wipes 4.7 Anti-static tools
5. Cleaning Methods	5.1 Dry wiping 5.2 Wet wiping 5.3 Compressed air dust removal 5.4 Anti-static cleaning
6. Safety	6.1 PPE (gloves, goggles, anti-static wrist straps) 6.2 Power-off equipment 6.3 Proper ventilation 6.4 Safe handling of chemicals
7. Diagnostic tools	7.1 Multimeters 7.2 Cable testers 7.3 System monitoring software 7.4 Event log analysers 7.5 Built-in OS diagnostics 7.6 Third-party diagnostic utilities
8. Software systems	8.1 Operating systems 8.2 Application software 8.3 Drivers 8.4 Firmware 8.5 Network management software
9. Documented	9.1 Diagnostic logs 9.2 Fault reports 9.3 Repair recommendations 9.4 Maintenance records
10. Components	10.1 Motherboards 10.2 CPUs 10.3 RAM 10.4 Hard drives/SSDs

	10.5 Power supplies 10.6 Network cards 10.7 Peripheral devices 10.8 Printers 10.9 Monitors
11. Repair methods	11.1 Soldering 11.2 Firmware updates 11.3 Component calibration 11.4 Cleaning contacts 11.5 Software reinstallation 11.6 Configuration correction
12. Replacement parts	12.1 Original manufacturer parts 12.2 Compatible spares 12.3 Network device modules 12.4 Storage drives
13. Updates	13.1 Operating system updates 13.2 Application updates 13.3 Firmware updates 13.4 Security patches 13.5 Driver updates
14. Installation tools	14.1 OS update utilities 14.2 Device manager 14.3 Vendor-provided installation packages 14.4 Command-line tools 14.5 Patch management software
15. Testing tools	15.1 Diagnostic software 15.2 System monitoring tools 15.3 Stress testing utilities 15.4 Built-in OS test features 15.5 Performance benchmarking tools
16. Testing procedures	16.1 Functional testing 16.2 Performance testing 16.3 Connectivity testing 16.4 User acceptance testing 16.5 Load/stress testing
17. Expected outcomes	17.1 System stability 17.2 Correct functionality 17.3 Compliance with specifications 17.4 Successful integration
18. Issues	18.1 Hardware failures 18.2 Software errors 18.3 Configuration conflicts 18.4 Network downtime 18.5 Security breaches 18.6 Performance degradation
19. Solutions	19.1 Component repair/replacement 19.2 Software patching

	19.3 Configuration adjustments 19.4 System updates 19.5 Security fixes
20. Documentation	20.1 Incident reports 20.2 Troubleshooting logs 20.3 Service tickets 20.4 Knowledge base entries 20.5 Maintenance records

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify tools, equipment, and procedures required for inspecting hardware components. 1.2 Understand methods for detecting physical damage, electrical faults, and performance issues in hardware. 1.3 Interpret cleaning techniques and safety precautions for maintaining internal and external computer parts. 1.4 Recognize how to diagnose hardware and software issues using appropriate tools and software utilities. 1.5 Understand procedures for isolating, repairing, or replacing faulty components. 1.6 Identify the importance of installing correct updates, drivers, and patches. 1.7 Interpret standard system testing methods to verify functionality, performance, and reliability. 1.8 Understand documentation procedures for recording issues, root causes, and applied solutions. 1.9 Recognize the importance of maintaining accurate maintenance records.
2. Underpinning Skills	2.1 Performing inspection of hardware components to detect faults, damage, or malfunctions. 2.2 Selecting and preparing appropriate tools and equipment for cleaning and maintenance tasks. 2.3 Cleaning internal and external hardware components carefully following safety and manufacturer guidelines. 2.4 Diagnosing hardware and software issues using diagnostic tools and software utilities. 2.5 Repairing or replacing faulty components using approved procedures and tools. 2.6 Installing updates, drivers, and patches to maintain system functionality and compatibility. 2.7 Testing system components and software to verify proper operation, performance, and reliability. 2.8 Documenting issues, root causes, and corrective actions accurately for future reference.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns

	3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Relevant tools, Equipment, software and facilities needed to perform the activities. 4.3 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 inspected hardware components for faults. 1.2 cleaned internal and external parts. 1.3 diagnosed hardware and software issues. 1.4 repaired or replaced faulty components. 1.5 installed updates, drivers, and patches. 1.6 tested systems to ensure functionality. 1.7 documented issues and solutions for future reference.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	IT Support Service
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)
- OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptop	01
2.	Printer	01
3.	Scanner	01
4.	Computers/Laptop: (Minimum Intel Core i5 processor (8 th Gen), 8GB RAM (16GB preferred), 500GB SSD/HDD storage (1TB preferred), High-resolution monitors (19/20-inch minimum, with color accuracy).	25
5.	Projector: High-resolution projector (HD or 4K) for instructor presentations, large screen for the room to ensure visibility for all trainees.	01
6.	Network Devices (Switches and Wi-Fi Routers)	05 sets
7.	Network tools (Clamping tools and cable testers)	10 sets each
8.	Network Cables (UTP and Fiber Optics)	100 meters
9.	Connectors (RJ 45, MTP/MPO)	100
10.	Ergonomic chairs and desks	25
11.	Internet connectivity with power back-up	01

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on IT Support Service, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.